# Jimmy Nguyen

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## Education

**Georgia Institute of Technology** *Expected Jun 2028*
*Master of Science in Computer Science* | Specialization: Artificial Intelligence
Coursework:*Game AI, Introduction to Graduate Algorithms, Software Development Process*

**University of California Davis** *Jun 2024*
*Bachelor of Science in Computer Science* GPA: 3.7
Coursework:*Data Structures and Algorithms, Algorithm Design and Analysis, Operating Systems, Computer Architecture, Programming Languages, Computer Vision, Machine Learning, Artificial Intelligence*

## Work Experience

**Apple Career Experience** *Jun 2026 - Nov 2026*
Software Support Engineer Sunnyvale, CA

- **Designed and built a full-stack internal web application** (Python, **FastAPI**, PostgreSQL) used across retail locations to manage and provision hardware devices, replacing a manual process with **self-service**.
- **Implemented secure single sign-on** (OAuth/OIDC) with **role-based access control**, and deployed the app to a **Kubernetes**-based platform via automated **CI/CD pipelines** and GitOps-style delivery.
- **Diagnosed and resolved production reliability issues**, including a memory-exhaustion crash and a gateway-timeout failure, by introducing **concurrency controls** and infrastructure configuration fixes.
- **Built a Python-based QA automation framework** combining UI automation with **AI-driven visual validation** to test iOS/iPadOS applications across 10+ locales, **cutting manual regression testing time**.
- **Automated bug-triage workflows** by building a scheduled service that parsed structured internal bug reports and orchestrated backend actions automatically, and filed detailed, reproducible defect reports.

## Projects

**Pokémon Sealed Inventory Tracker** | React | Python | FastAPI | SQLite *Jan 2026 – Nov 2026*

- Built a **full-stack portfolio tracker** with **gain/loss analytics**, allocation breakdowns, and a **24-month value history** fed by an external pricing API, with the frontend hosted on **GitHub Pages** and the API on **Render**.
- Built a **FastAPI REST backend** on SQLite with a **price-history model** and Pydantic request validation.
- **Cut API latency up to 74%** (74 $\rightarrow$ 19 ms) at 10K holdings by moving aggregation and grouping into **SQL**.
- Added **product search**, custom categories, and purchase tracking, with a **responsive layout** for mobile.
- Reached **99% backend / 90% frontend test coverage** (46 tests), enforced in **GitHub Actions** CI.
- Designed a **React** dashboard with hand-built SVG charts, scoring **Lighthouse 100** in all four categories.

**Credit Score Explorer** | Python | scikit-learn | Streamlit | pandas *Mar 2025 – Nov 2026*

- **Led a team of 5** to build a credit score classifier on 100K financial records, later **rebuilt** as a deployed, **explainable ML web application** in Streamlit with a fully reproducible training and evaluation pipeline.
- Designed an **ordinal gradient-boosting model with monotonic constraints**, so more debt can never improve a score, plus a **what-if search** that finds the smallest change to raise a user's predicted score band.
- Built **model-agnostic explanations** showing which factors drive each user's predicted score band.
- Showed actionable what-if results in-app, e.g., cutting debt from $3,400 to $1,200 to raise a band.
- Achieved **66% accuracy** (53% baseline) on **held-out customers** via a leakage-free, customer-level split.

## Technical Skills

**Languages:** Python, C/C++, JavaScript, TypeScript, HTML/CSS, Java, Erlang, Matlab

**Tools & Technologies:** React, FastAPI, scikit-learn, TensorFlow, pandas, Streamlit, SQLite, PostgreSQL, Git, Docker, Kubernetes, REST APIs, CI/CD (GitHub Actions), OAuth/OIDC, Playwright, Agile/Scrum workflows